

SA-RMAC: A Sparsity-Aware INT8 Neural Network Accelerator with Reconfigurable MAC Array

VLSID 2027 | User Design Track

Motivation: dense MAC arrays waste work and energy on zeros

Context

- Edge and embedded AI needs low-latency, low-power DNN inference.
- DNN activation functions (e.g., ReLU) produce many zero activations, so many MACs do no useful work.

Limitations

- **Dense** accelerators multiply by zero and keep every MAC clocked.
- Fixed-width MAC arrays cannot adapt to light workloads.

Why it matters

- Lightweight DNN inference should fire only as many MACs as the data needs.
- Small teams need reusable AI IP, verified against the **CMSIS-NN** INT8 reference, for FPGA and ASIC flows.

Example input vector
8 × INT8 values

0	7	0	-3	0	5	0	2

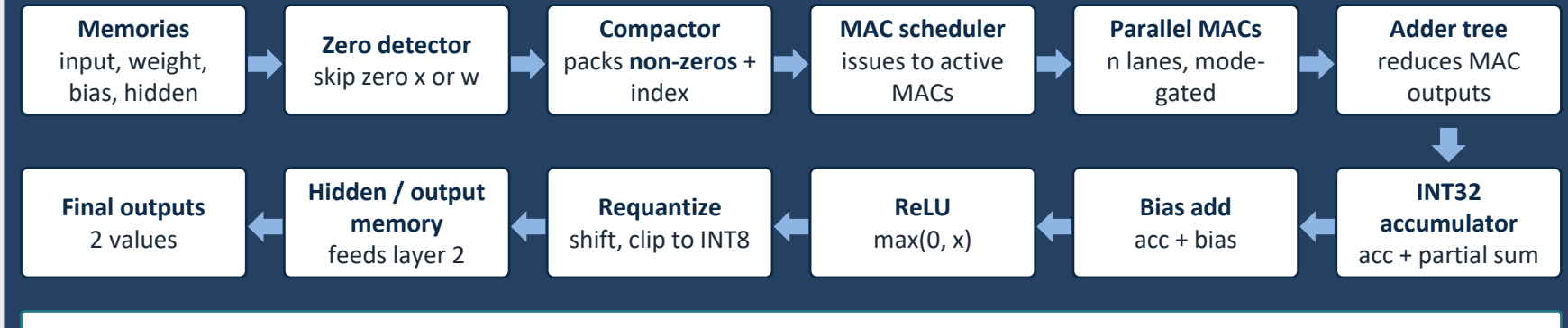
4 of 8 MACs
do no useful work

80 MACs per inference for an 8→8→2 DNN (MLP): 8×8 in the hidden layer plus 8×2 in the output layer. A small workload where zero-skipping and power modes can be studied exactly.

Main idea: skip the zeros, power only the MACs needed

Control FSM: IDLE → LOAD → MAC → BIAS → RELU → STORE → DONE (MAC state runs until all non-zero pairs are consumed)

ARCHITECTURE: DNN INFERENCE DATAPATH



Power-aware mode controller: non-zero count → 1 / 4 / 8 active MACs; idle MACs clock-gated

Sparsity-aware

Cycles per neuron = max(1, ⌈non-zeros / N⌉)

for N active MACs.

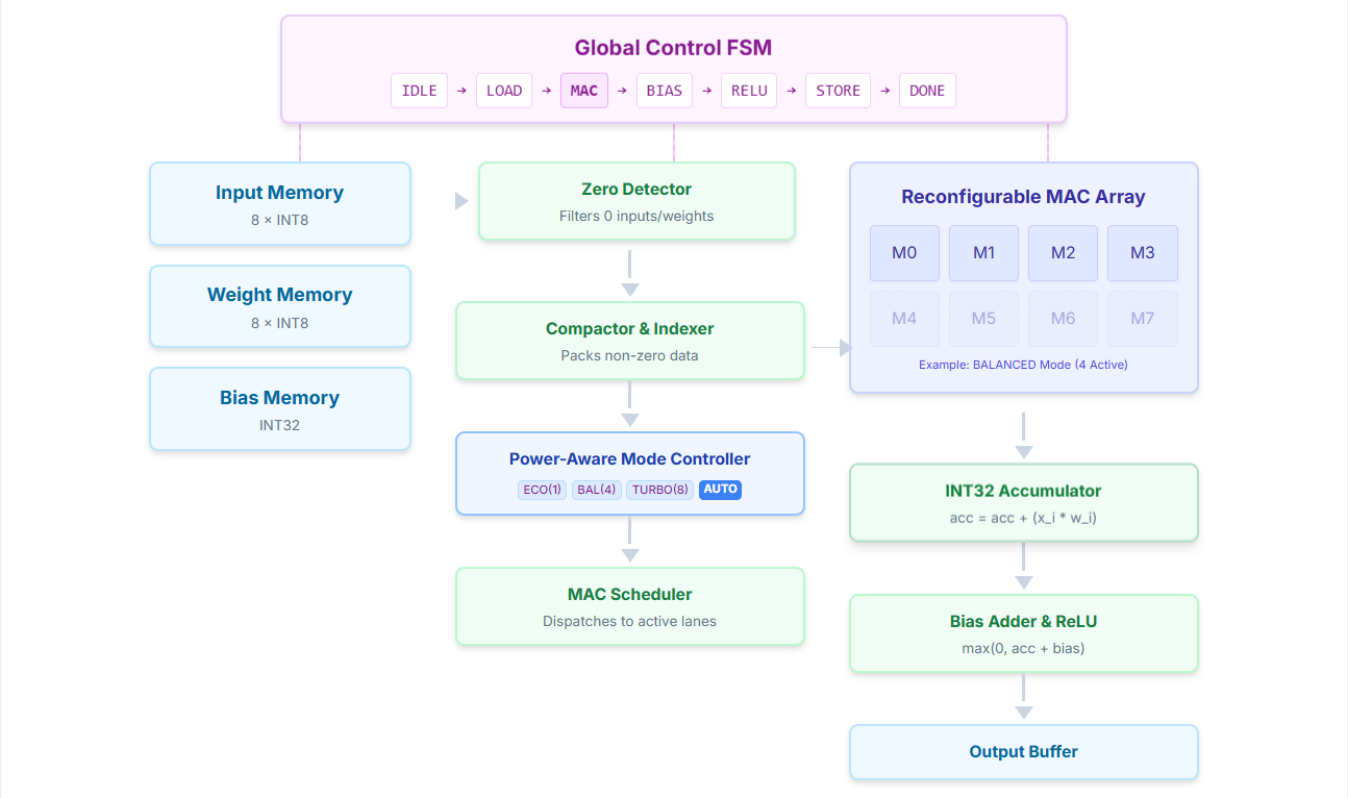
Power-aware

ECO, **BALANCED**, **TURBO** or **AUTO** sets N per neuron; idle MACs are **clock-gated**.

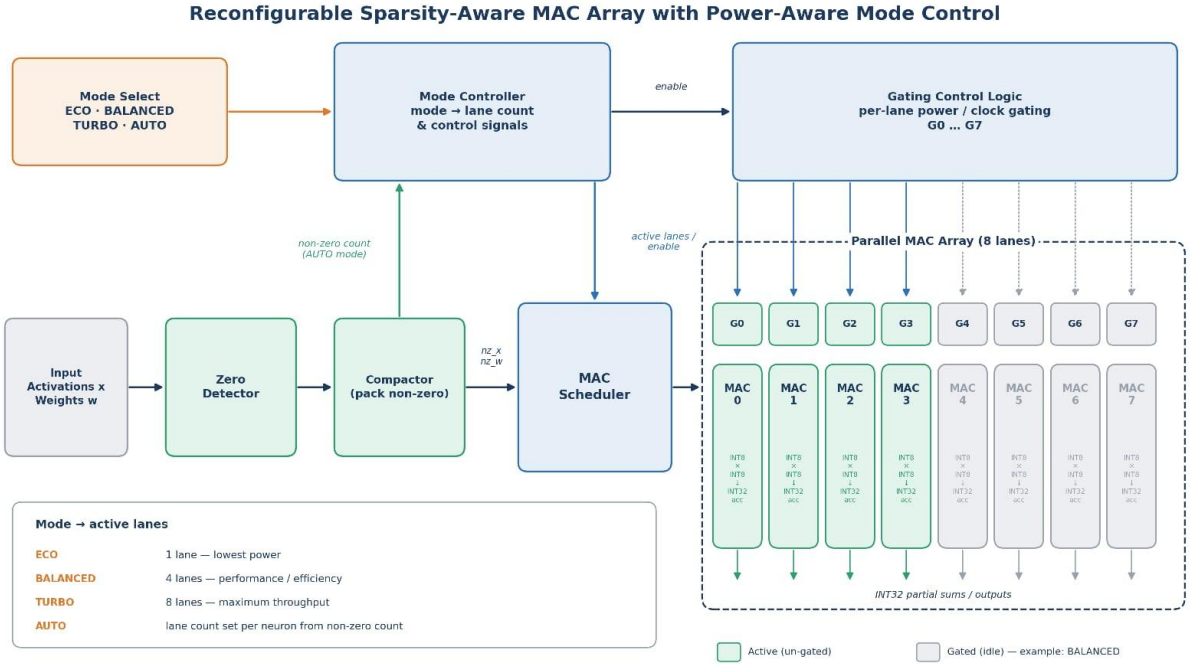
Verified flow

Compared with the **CMSIS-NN INT8 reference kernel**; self-checking RTL testbench in dense, sparse and every power mode.

SA-RMAC Overall Architecture



Reconfigurable MAC array: mode control and lane-level gating



Architecture Insights

- **Mode select** sets the lane count: ECO 1, BALANCED 4, TURBO 8; AUTO uses the non-zero count.
- **Zero detector** and compactor pack non-zero (x, w) pairs for the MAC scheduler.
- **Gating logic** powers only the needed lanes; idle lanes are clock-gated.
- Shown: BALANCED, with lanes 0 to 3 active and 4 to 7 gated.

Dynamic Mode Configuration

ECO Mode

Ultra-low power. 1 active MAC. Up to 8 cycles/neuron.

Active Lanes: 1 / 8

MAC 0	MAC 1
MAC 2	MAC 3
MAC 4	MAC 5
MAC 6	MAC 7

OPTIMAL AT 50% SPARSITY

BALANCED Mode

Energy-delay optimal. 4 active MACs. Up to 2 cycles/neuron.

Active Lanes: 4 / 8

MAC 0	MAC 1
MAC 2	MAC 3
MAC 4	MAC 5
MAC 6	MAC 7

TURBO Mode

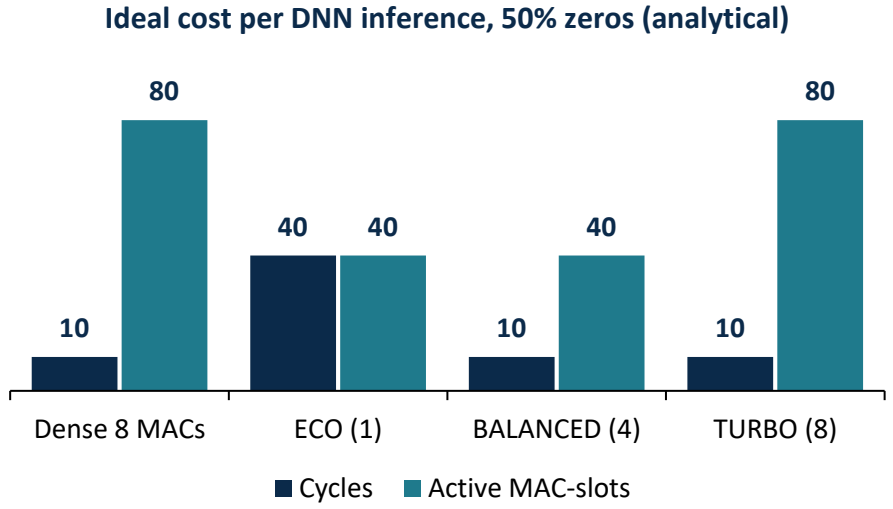
Max throughput. 8 active MACs. 1 cycle/neuron.

Active Lanes: 8 / 8

MAC 0	MAC 1
MAC 2	MAC 3
MAC 4	MAC 5
MAC 6	MAC 7

AUTO Mode: Dynamically calculates `ceil(non_zero_count / N)` to select the minimum power mode that completes the neuron computation in a single cycle.

Trade-offs: latency vs. active hardware across power modes



Architectural insights

- Skipping zeros keeps latency at **10 cycles** (MAC state only) for 4 and 8 MACs; **ECO** stretches to 40.
- At **50% zeros**, **BALANCED** halves active MAC-slots versus **TURBO** (**40 vs 80**) at equal latency.
- **AUTO** picks the smallest group that finishes a neuron in one cycle, so it wins when **non-zero** counts vary by neuron or layer.
- Detector, compactor and mode controller add area to the DNN datapath; to be quantified from Vivado utilisation reports.

ECO · 1 MAC

Lowest activity; up to 8 cycles per neuron. For light, low-rate work.

BALANCED · 4 MACs

Up to 2 cycles per neuron. Good speed and activity trade-off.

TURBO · 8 MACs

1 cycle per neuron. Highest throughput, most MACs active.

Proposed improvements across power modes

Metric	ECO	BAL.	TURBO	AUTO
Cycles, 0% zeros	80	20	10	10
Cycles, 50% zeros	40	10	10	10
Cycles, 75% zeros	20	10	10	10
MAC-slots, 50% zeros	40	40	80	40
MAC-slots, 75% zeros	20	40	80	40
Slot reduction vs dense, 50%	2.0×	2.0×	1.0×	2.0×
Measured: cycles, skip %	*	*	*	*
Measured: mW, Fmax, LUTs	*	*	*	*

Verification vs. CMSIS-NN reference

- Random, positive, negative, zero and boundary (−128, 127) vectors
- Sparsity swept: 0%, 50%, 75%, 100% zeros
- Dense, sparse and every power mode vs. **CMSIS-NN arm_fully_connected_s8: 0 mismatches**
- Reference follows the TFLite Micro INT8 quantization spec

Measurement flow

- Simulation: cycle and skipped-MAC counters
- Vivado: utilisation, Fmax, power estimate from switching activity
- FPGA board: functional check

** These values are to be calculated from RTL simulation and Vivado reports.*

Summary

Key contributions

- Zero detection, compaction and scheduling of **non-zero** pairs.
- **Power-aware** modes: **ECO**, **BALANCED**, **TURBO** and **AUTO**.
- RTL-to-FPGA flow **verified against the CMSIS-NN INT8 reference** kernel.

Results

- At **50% zeros**, **BALANCED** matches the 10-cycle dense 8-MAC latency with half the active MAC-slots (**40 vs 80**, analytical).
- Measured cycles, power, Fmax and utilisation are reported in the results table.

Impact

- Lower switching activity for lightweight edge DNN inference.
- Reusable, verified IP that scales to larger DNNs.

Reference

- CMSIS_NN_software library

Application

- Wearable & Portable Electronics
- Energy efficient computation in DNN and other accelerator.
- Industrial Edge AI
- Smart Cameras & Vision Systems

Future scope

RISC-V memory-mapped interface | **Larger DNNs, layer tiling** | **Pipelining for Fmax** | **ASIC flow**

Takeaway: SA-RMAC skips the zeros, powers only the MACs a DNN layer needs and is verified against the CMSIS-NN INT8 reference kernel.